

Project Kickoff Report v1

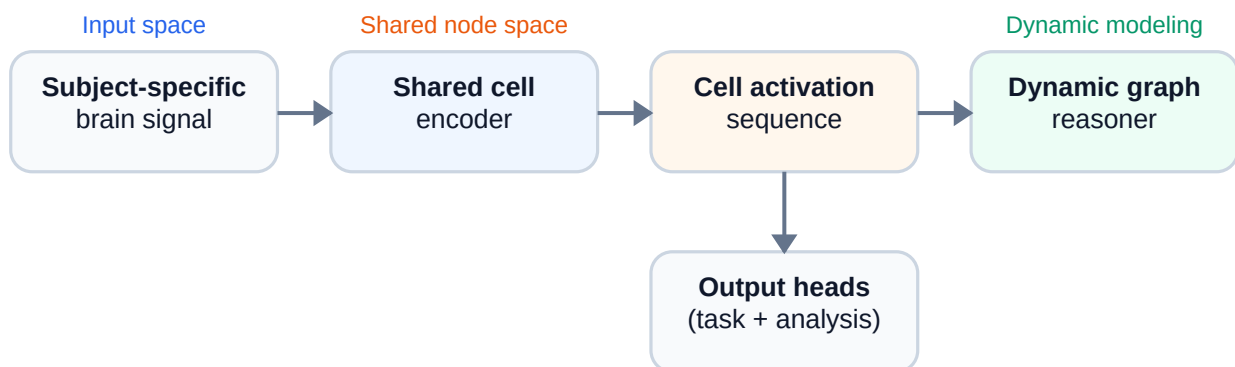
Problem-to-Pipeline startup report for a cross-subject visual brain graph learning project

This report translates the current proposal into a first workable research program. The immediate goal is not to build the final dynamic GNN, but to fix the problem definition, the dataset strategy, the graph object, the operational meaning of shared perceptual cells, and a minimal viable experiment (MVE) that can be started within 1-2 months.

Current first-phase recommendation

Use public visual **fMRI** as the primary modality; use **NSD** for method development and **THINGS/BOLD5000** for external validation. Start with a **hybrid graph**: subject-specific signals -> shared perceptual cells -> graph reasoning at the cell level. First validate the **shared cell space** before adding full dynamic graph modeling.

Overall pipeline sketch



The sequencing principle is: **define the shared node space first, then test whether graph reasoning adds value on top of it**. Otherwise any gain from a dynamic GNN could be confounded with better node alignment or a stronger non-graph temporal model.

1. Research question concretization

1.1 Main problem

Given multi-subject brain responses during visual tasks, can we learn a **cross-subject shared mesoscopic node space** (shared perceptual cells) that supports downstream graph modeling better than fixed ROI graphs or subject-specific latent spaces? On top of this node space, can we model **coarse time-varying functional interactions** that are informative for visual category or recognition-related tasks? Generalization here means performance and representational stability on **held-out subjects**, not only random-trial accuracy.

1.2 Three subquestions

- **Q1 - Shared representation.** Can we learn a shared perceptual cell space from multiple subjects such that samples become more comparable across subjects than under subject-specific latent spaces or fixed ROI representations?
- **Q2 - Dynamic interaction.** Once a shared node space exists, do there exist meaningful task-related interaction patterns on that space, and can they be modeled more effectively as graphs than as non-graph temporal summaries?
- **Q3 - Generalization.** Does the shared node space, with or without graph reasoning, improve held-out-subject generalization, low-shot adaptation, and external validation on another public visual fMRI dataset?

1.3 Non-goals for the first phase

- No attempt to recover neuron-level mechanisms or millisecond neural transmission order.
- No claim of full-brain causal discovery from fMRI.
- No requirement to jointly train fMRI and sEEG in Phase 1.
- No requirement to reach state-of-the-art decoding performance as the first milestone.
- No full dynamic graph generation model before the shared node space is validated.

1.4 Why this problem matters

Current brain-graph pipelines often assume a fixed atlas-defined node space. That assumption is convenient but brittle for cross-subject learning, because the real difficulty is not only graph reasoning but also the definition of **which nodes should be shared across subjects**. A successful first version would establish whether a learned node space is a more useful foundation than fixed ROIs for later dynamic graph modeling and transfer.

2. Dataset and signal inventory

The dataset decision should be driven by what the first experiment must prove: a shared node space should be testable under a clean visual-task setting with usable cross-subject structure. This strongly favors public visual fMRI over sEEG as the Phase 1 main source.

Dataset	Modality	Subjects	Task / labels	Temporal / spatial	Cross-subject suitability	Role in project
NSD	7T fMRI	8	Natural scenes + continuous recognition	1.8 mm	High: same modality, deep penetration	Primary Phase 1 development
THINGS-fMRI	fMRI	3	Object images; category / concept labels	event-based; high-resolution	Moderate: fewer subjects, strong generalization	Generalization / object-agnostic
BOLD5000	fMRI	4	Natural images; image/category labels	event-related	Moderate: useful external checks	External validation
sEEG (generic cascade)	sEEG	typically small	often task-specific; coverage varies	ms; sparse spatial coverage	High for Phase 1 shared-space training	Later-stage extension / validation

Decision. Use **NSD** as the main Phase 1 dataset; use **THINGS-fMRI** and/or **BOLD5000** as external validation. Treat sEEG as a later-stage extension because it offers excellent temporal precision but does not provide a comparably convenient main training regime for cross-subject visual graph learning.

2.1 Why fMRI first is still defensible despite low temporal resolution

fMRI cannot support fine neural-timescale interaction claims for visual processing, but it can support **coarse event-related graph-state evolution** over a few broad bins. The objective is not to recover millisecond propagation, but to model **stimulus-conditioned changes in coarse mesoscopic graph states** and test whether such states generalize across subjects.

2.2 Suitability conclusion

The comparison above supports a clean sequencing decision: (i) build the shared node space on public visual fMRI; (ii) test whether graph reasoning adds value on that space; and only later (iii) revisit higher-temporal-resolution modalities for finer dynamic claims.

3. Graph construction design choices

The central design question is not only which GNN to use, but **what graph object should exist at all**. The kickoff stage compares three candidate graph definitions.

Option	Definition	Interpretability	Alignment feasibility	Flexibility	Implementation risk
A. ROI graph	Nodes = atlas ROIs; edges = FC / correlation / learned relationships	High	High	Moderate	Low
B. Latent cell graph	Nodes = learned perceptual cells; edges = cell-cell interactions	Moderate if stable	Potentially high	High	High
C. Hybrid graph	Subject signals -> shared cells -> graph reasoning at cell level	Moderate-high	High if cells are stable	High	Moderate

3.1 Analysis of the three options

- **ROI graph.** Best for ease, baseline construction, and interpretability, but it undercuts the core research question because the node space is fixed by atlas design rather than learned from data.
- **Latent cell graph.** Best aligned with the proposal's ambition, but too risky as a first version because both the node space and the graph dynamics would be learned simultaneously from scratch.
- **Hybrid graph.** Best trade-off for the kickoff phase: it allows node learning to be the main innovation while keeping later graph reasoning technically modular.

3.2 First-phase decision

The recommended first-phase design is the **hybrid graph**. Raw or pre-aggregated subject-specific brain signals are encoded into shared perceptual cells; the cell activations become node features; graph reasoning, if used, is performed on the cell graph. This preserves the proposal's novelty without forcing the first experiment to solve the entire dynamic-graph problem at once.

4. Shared perceptual cells: operationalization

A shared perceptual cell should be defined operationally, not rhetorically. In this project, a perceptual cell is a **shared mesoscopic latent prototype** that can absorb subject-specific signal and produce a comparable activation across subjects for the same or similar stimuli.

4.1 Operational definition

- A cell is not a neuron and not necessarily a fixed anatomical parcel.
- A cell is a shared latent prototype or basis element used as a candidate graph node.
- The number of cells is fixed during training for a chosen model instance (e.g., 32, 64, or 100).
- Cells interact with subject-specific signals through an assignment or projection mechanism.
- Their usefulness is judged by cross-subject comparability, task utility, and stability.

Route	Core idea	End-to-end?	Cross-subject sharing	Interpretability	Fit for later graph use
Prototype/token bank	Learn K shared prototypes and assign signals to them	Yes	Strong	Good if stable	Excellent
Clustering-derived cells	Cluster learned representations into cells	Usually no	Moderate	Moderate	Fair
Dictionary/factorization	Shared low-rank basis over subject signals	Sometimes	Strong	Good	Good

4.2 First implementation decision

The first implementation should use a **prototype/token-based shared cell bank**. This choice best supports end-to-end training, explicit assignment, and a clean handoff to later graph reasoning. It also produces the most direct answer to Q1: whether a shared latent node space can be learned at all.

5. Minimal viable experiment (MVE)

The MVE should test one core claim only: **a shared cell space improves cross-subject generalization relative to subject-specific or fixed-space baselines**. Full dynamic GNN modeling is deferred until this claim is validated.

Component	First workable version
Input	Trial- or stimulus-level fMRI representation from a selected visual dataset (NSD first)
Encoder	Subject-specific encoder -> shared prototype bank -> cell activations
Task	Cross-subject category decoding and/or cross-subject retrieval
Baselines	Subject-specific MLP; shared linear latent space; ROI-based model without learned cells
Loss	Task loss + alignment regularization + prototype diversity / collapse prevention
Primary metrics	Held-out-subject accuracy / retrieval Recall@K / RSA similarity / cell consistency
Success criterion	Shared-cell model beats subject-specific and simple shared-space baselines on held-out subjects

5.1 Why the MVE intentionally avoids full dynamic GNN

If a dynamic graph model is added before the shared node space is validated, it becomes impossible to tell whether improvements come from better node alignment, better temporal modeling, or simply more parameters. The MVE therefore isolates the node-space question first. Only after that should Version 1 add static or weak graph interaction, and Version 2 add a genuine dynamic GNN.

5.2 Planned baselines and future ablations

- Baselines must include a strong non-graph model on the same shared node space.
- Ablations should compare fixed ROI vs learned cells, and scaffold-only vs learned graph edges once graph reasoning is introduced.
- A later success condition is that graph reasoning must add value beyond a strong temporal model on the same shared nodes.

6. Risk list and decision points

Risk	Why it may happen	Detection signal	Fallback
Shared cells collapse	Prototype assignment becomes unbalanced	Low diversity / identical cells	Entropy/diversity regularization; reduce K
Cells are mathematically well-behaved but empirically weak	Model is too simple	Poor cross-subject consistency	Add alignment / retrieval objective
Dynamic graph is unstable	Edge freedom too high for data regime	Edge patterns vary wildly across subjects	Use scaffold-only or weak dynamic edges first
fMRI temporal claim is too strong	Model is slow	Reviewer criticism; weak interpretation	Relax claim to coarse graph-state evolution
External validation fails	Dataset-specific bias	Strong NSD, weak external set	Tighten regularization; simplify task
Too many moving parts	Scope inflation	Slow progress across all fronts	Freeze Version 0 before graph Version 1
sEEG distracts the main theme	Interesting but misaligned regime	No reproducible Phase 1 result	Keep sEEG for later extension only

6.1 Immediate decision points

- **D1.** Select the first-phase main dataset: NSD versus another public visual fMRI dataset.
- **D2.** Fix the first implementation of shared cells: prototype/token bank versus another route.
- **D3.** Fix the primary metric of the MVE: held-out-subject classification, retrieval, or RSA-driven evaluation.

6.2 Four-week execution plan

Week	Deliverables	Potential blocker
1	Literature matrix; dataset inventory; problem statement; pipeline sketch	Problem framing still too broad
2	Graph-choice comparison; shared-cell route decision; MVE spec	Graph object and baseline set not settled
3	Kickoff report draft; risk register; glossary; data-flow table	Sections not yet coherent
4	Final PDF report; kickoff assets; executive summary; short lab presentation	MVE success criterion still vague

6.3 Final recommendation

The project should officially start with **Version 0**: validate the shared cell space on public visual fMRI. Only after Version 0 yields a stable positive result should the project advance to graph interaction and then to full dynamic graph reasoning.

Selected references

- [R1] Natural Scenes Dataset official site: 8 subjects, 1.8-mm isotropic fMRI, 1.6-s sampling rate, 30-40 sessions.
- [R2] THINGS-fMRI dataset overview: 3 participants, 12 sessions, 8,740 unique images of 720 object concepts.
- [R3] BOLD5000 official site: 4 participants, 5,254 images in a slow event-related fMRI design.
- [R4] Xu et al. Inductive Representation Learning on Temporal Graphs (TGAT), ICLR 2020.
- [R5] Trivedi et al. DyRep: Learning Representations over Dynamic Graphs, ICLR 2019.
- [R6] Li et al. BrainGNN: Interpretable Brain Graph Neural Network for fMRI Analysis, Medical Image Analysis 2021.